Choroidal Melanoma Treatment With Iodine 125 Brachytherapy

Bruce R. Garretson, MD; Dennis M. Robertson, MD; John D. Earle, MD

• Twenty-six patients were treated for malignant melanoma of the choroid with iodine 125 episcleral plaques. Patients were followed up for at least two years, with a mean follow-up of 45 months. Tumor shrinkage was documented by A-scan ultrasonography in all but one patient. The average reduction in tumor thickness was 46%. Visual acuity remained within two Snellen lines of preoperative levels in 14 (54%) of the 26 patients. Of the 12 patients who lost more than two lines of central visual acuity, eight developed radiation changes affecting either the optic nerve or retina, and three developed cataracts; enucleation accounted for the loss of vision in the final patient. Enucleation was necessary in one other patient in this series because of tumor growth. The mean duration between treatment with iodine 125 and the development of radiation changes affecting the optic nerve or retina was 32 months.

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Although the treatment of malignant melanoma of the choroid remains controversial, radiation therapy has become an increasingly popular alternative to enucleation. Its origin dates to the use of radon by Moore[1] in 1930. Various forms of radiation have been used in different centers, including cobalt 60,[2] ruthenium,[3] helium ion,[4] proton beam,[5] iodine 125,[6,7] iridium 192,[8] and gold 198.[9] Each type of radiation has its advocates, although radiation complications have been significant in many series.[10-12]

Although most of the American experience in brachytherapy for uveal melanoma has been with cobalt 60, in recent years, the isotope iodine 125 has been considered by some investigators to be a more desirable radiation source because of its advantageous physical characteristics, which result in the singular advantage that its radiation can be almost totally shielded with a thin (approximately 0.5-mm thick) plate of gold. We take advantage of this shielding capacity by designing the iodine 125 therapeutic plaque with a thin gold backing. Since the radiation from iodine 125 can essentially be totally shielded, the potential radiation hazard to persons handling the plaque is minimized. Also, a preferential directional radiation pattern can be concentrated toward the tumor in a direct line with the unshielded iodine 125, while at the same time shielding tissues behind the plaque. Because of the real and theoretical advantages of iodine 125, proponents of its use have been hopeful that the incidence of radiation retinopathy and optic neuropathy will be lessened when compared with other radiation sources or that at least the onset of these complications will be delayed. Since there is little written on the clinical use of iodine 125 brachytherapy, except for publications reporting preliminary observations,[6,7] we herein report our clinical experience with iodine 125 brachytherapy in the management of choroidal melanoma in a small series of patients followed up for two or more years.

PATIENTS AND METHODS

Twenty-six patients with the clinical diagnosis of choroidal melanoma were treated with iodine 125 episcleral plaques between Sept 12, 1980, and July 10, 1984. The preoperative evaluation included a complete ophthalmic examination, fundus drawings, color and fluorescein angiographic photography of the fundus, and A- and B-scan ultrasonography. All patients included were free of metastatic disease, as determined by systemic medical and radiologic evaluation prior to inclusion in the study. The evaluation included a general physical examination; laboratory tests, including serum aspartate aminotransferase, serum alanine aminotransferase, serum γ-glutamyl transferase, lactate dehydrogenase, and alkaline phosphatase determinations; chest roentgenogram; and abdominal computed tomographic scanning.

After estimation of tumor-base dimension and height were finalized, a cup-shaped plaque composed of a 0.4-mm thick gold plate was fashioned so as to encompass the base of the tumor and a 2-mm tumor-free perimeter (detailed preparation previously reported[13]). With knowledge of tumor size, computer-generated point dose calculation methods, utilizing the ARTRONIX (Artronix Inc, St Louis) or AECL (Atomic Energy of Canada, Limited; Kanata, Ontario) treatment planning computers, were performed to determine the number of iodine 125 seeds needed to deliver approximately 100 Gy (10 000 rad) to the tumor apex over a five- to six-day exposure interval. The hourly dose was calculated to be 0.5 to 1.0 Gy (50 to 100 rad) at the tumor apex. (Note: The first two tumors received approximately 80 Gy [8000 rad] to the apex.)

All plaques were placed on the scleral surface with the patient under general anesthesia. After performing a conjunctival limbal peritomy, the tumor margins were identified and marked on the sclera, utilizing standard techniques for localizing retinal breaks, transillumination, or both. The therapeutic plaque was centered over the tumor base and anchored with two to three partial-thickness scleral sutures that were passed through positioning holes fashioned in the plaque. Recti muscles were disinserted as necessary to facilitate placement of the plaque. The surgical bed was irrigated with antibiotic solution prior to closing Tenon's capsule and conjunctiva. After the appropriate interval for radiation delivery was completed, the plaques

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From the Departments of Ophthalmology (Drs Garretson and Robertson) and Radiation Therapy (Dr Earle), Mayo Clinic, Rochester, Minn.

Reprint requests to the Department of Ophthalmology, Mayo Clinic, Rochester, MN 55905 (Dr Robertson).

Choroidal Melanoma and Iodine 125 Brachytherapy*								
Patient No./ Age, y/ Sex	Eye	Size of Tumor	Distance, Disc Diameters†		Visual Acuity		Follow-up Duration, mo	Comment
			Fovea	Optic Nerve	Preoperative	Postoperative		
1/36/M	OD	ME	7	5	20/20	NLP	76	VA 20/25 for 50 months; traction retinal detachment secondary to radiation; enucleation, January 1986
2/60/F	OS	ME	3	3	20/20	CF	63	VA 20/20 for 39 mo; CME and optic atrophy
3/64/M	OS	ME	5	3	20/20	20/25	71	No retinopathy
4/62/M	OS	ME	3	5	20/50	20/400	58	Nuclear cataract
5/57/M	OD	ME	0	1	20/200	20/70+	33	Died of adenocarcinoma of lung
6/67/F	OS	ME	>5	>5	20/20	20/20	57	No retinopathy
7/67/M	OS	ME	>5	>5	20/20	20/20	54	Mild PSC change
8/46/M	OS	S	0	2	20/400	20/200	54	Radiation maculopathy at 26 mo
9/51/M	OS	L	1.5	3.5	20/20	20/30	52	Mild CME
10/57/M	OD	L	>5	>5	20/40	HM	52	Mature PSC cataract
11/58/F	OD	L	2	1.5	20/20	20/100	50	PSC cataract
12/73/M	OD	ME	0	1	20/100	2/200	49	Radiation nerve damage at 19 mo
13/70/M	OD	L	4.5	3	20/25	20/60	48	Branch retinal artery occlusion and one PSC at 43 mo
14/67/F	OD	S	3	4	20/25	NLP	45	Enucleation at 6 mo due to tumor growth
15/71/F	OD	L	3.5	1	20/25	20/300	42	Macular ischemia at 37 mo
16/53/F	OS	ME	4	2	20/30	20/20	41	Resolution of serous retinal detachment
17/62/M	OS	L	>5	>5	20/20	20/30	39	Mild PSC cataracts
18/63/F	OS	L	3	3.5	20/100	20/400	39	Macular hole at presentation
19/40/F	OD	ME	2	4	20/20	20/30	39	No retinopathy
20/71/F	OD	ME	4	4.5	20/20	20/20	36	No retinopathy
21/54/M	OS	L	3	4	20/25	20/200	36	CME
22/43/F	OS	ME	5	3.5	20/20	20/20	36	No retinopathy
23/20/M	OS	ME	3	2	20/100	20/300	30	Persistent serous retinal detachment
24/71/F	OS	ME	8	6	20/60	20/30	21	Died of metastatic melanoma
25/71/F	OD	L	3.5	1.5	20/40	20/400	26	CME
26/77/M	OS	S	0.75	1.5	20/70	20/100	26	Optic neuropathy

*S indicates small; ME, medium; L, large; NLP, no light perception; CF, counting fingers; HM, hand motions; VA, visual acuity; CME, cystoid macular edema; and PSC, posterior subcapsular cataract.

†One disc diameter equals 1.5 mm.

were removed, with the patient under local anesthesia, and any disinserted recti muscles were repositioned.

Follow-up examinations were generally performed at one month, three months, six months, one year, 18 months, and two years, and at six- to 12-month intervals thereafter. Examinations included visual acuity determination; complete ophthalmic examination, with fundus drawing and estimation of tumor-base dimensions; contact A- and B-scan ultrasonography, utilizing both the Ocuscan 400 (Sonometrics Systems Inc, New York) and the Bronson-Turner Ophthalmic "B" scan units (Grumman Health Systems; Woodbury, NY); and periodic fundus photography. Fluorescein angiography was performed in all patients at the one-year follow-up visit and yearly thereafter. Annual evaluation for metastatic disease included a chest roentgenogram, serum tests measuring liver function, and abdominal computed tomographic scan if indicated by abnormal results of liver function tests.

RESULTS

Patient Population

Of the 26 patients included in this series, there were 14 male and 12 female patients. Ages ranged from 20 to 77 years at surgery, with a mean age of 58 years. The right eye was involved in 11 patients and the left eye in 15. Tumor size at presentation was classified as small, medium, or large as follows: small, largest diameter less than 10 mm and height less than 3 mm; medium, largest diameter 10 mm or greater but less than 15 mm and height 3 mm or greater but less than 5 mm; or large, largest diameter 15 mm or greater and height 5 mm or greater. Our series contained three small, 14 medium, and nine large tumors (Table).

With the exception of one patient who died 21 months after treatment, all patients were followed up for at least two years, with a maximum follow-up of six years. The mean duration of follow-up was 45 months.

Eleven patients had tumors within 2 disc diameters (3 mm) of the fovea and/or optic nerve. Seven patients had tumors located between 2 and 3 disc diameters, from the fovea and/or optic nerve, and the remaining eight patients' tumors were farther than 3 disc diameters away. In total, 22 of the 26 patients had lesions within 5 disc diameters (7.5 mm) of the fovea and/or optic nerve. Localized serous retinal detachment was present in 14 of 26 patients, and in five of these, the fovea was involved.

Tumor Response

Reduction in tumor size was documented by A-scan ultrasonography and fundus photography in all but one patient in whom growth was documented six months following irradiation. The average reduction in tumor height, as measured by A-scan ultrasonography, was 48%. Small tumors decreased by an average of 66%; medium tumors, 44%; and large tumors, 49%.

Vision

Preoperative visual acuities ranged from 20/20 to 20/400. The visual acuity at the last follow-up examination was maintained within two Snellen lines of preoperative levels in 14 (54%) of 26 patients. Visual loss of greater than two lines occurred in the remaining 12 patients as follows: Significant cataract developed in three patients (two posterior subcapsular and one nuclear sclerotic), radiation retinopathy and/or optic neuropathy occurred in eight patients, and enucleation was performed in two patients. One patient whose eye was one of the eight with radiation retinopathy and/or optic neuropathy required enucleation because of radiation-induced complications. In the other patient, enucleation was performed because of tumor growth.

Vision was more likely to be affected when the treated tumor was located within 3 disc diameters of the optic nerve, fovea, or both. Ten (55%) of 18 such patients developed greater than two Snellen lines of visual acuity loss. In one patient, the eye was enucleated because of tumor growth, cataract developed in two patients (one nuclear and one posterior subcapsular), and radiation damage to the nerve and/or macula occurred in the remaining seven patients. Of the eight patients whose treated tumors were located farther than 3 disc diameters from the optic nerve and fovea, six (75%) retained visual acuities within two Snellen lines of pretreatment measurements. Of the other two patients, one retained 20/25 visual acuity for 50 months prior to the onset of radiation-induced complications that led to a total retinal detachment; the other developed a dense posterior subcapsular cataract.

In our series, radiation retinopathy and/or optic neuropathy causing greater than two lines of visual acuity loss occurred in eight (31%) of 26 patients. The mean interval between treatment and development of the retinopathy and/or neuropathy was 32 months (range, 12 to 50 months). Of these patients, cystoid macular edema accounted for visual loss in three patients, and macular ischemic changes occurred in two patients (in one patient, visual loss occurred from a branch retinal artery occlusion). Optic neuropathy was the cause of visual loss in two patients, and a progressive proliferative vitreoretinopathy, leading to total retinal detachment, developed in one patient prior to removal of the eye.

Retinal neovascularization and vitreous hemorrhage have not occurred in our series to date. Of the three patients who developed significant cataract, two developed cataracts that were of the posterior subcapsular variety and believed to be secondary to irradiation. The other patient developed progressive nuclear sclerotic changes in both eyes.

Other Major Complications

Enucleation was performed in two patients. The first patient retained 20/25 visual acuity for 50 months before the development of radiation retinopathy. Although the tumor had responded to radiation with shrinkage from a thickness of 3.5 to 2.0 mm, radiation complications, including progressive proliferative vitreoretinopathy leading to total retinal detachment and glaucoma, prompted enucleation 64 months after irradiation. The histologic evaluation of this eye revealed what appeared to be some nests of viable tumor cells in the residual tumor mass, yet the patient is alive, without evidence of metastatic disease at 76 months after irradiation. The second patient had documented tumor growth at six months following brachytherapy, and enucleation was performed. The tumor had increased in thickness from 3 to 6 mm, with collar-button formation. Histologic evaluation revealed a mixed-type choroidal melanoma, with focal areas of cellular necrosis interspersed with areas of what appeared to be viable tumor cells. Epithelial cell nuclear pyknosis and cytoplasmic ballooning degeneration were prominent in cells having epithelioid characteristics.

Two patients died during the follow-up period. One patient had biopsy-proved metastatic melanoma to the liver and died 21 months after plaque placement. The other patient died of histologically proved metastatic adenocarcinoma of the lung 33 months after treatment. Both originally presented with medium-sized melanomas, and in both, the melanomas had shown a response to radiation characterized by a 75% decrease in thickness in the first and a 30% decrease in thickness in the second.

COMMENT

The patient with malignant melanoma of the choroid presents a distinct challenge to the clinician with regard to selection of the proper course of management. Much confusion has arisen over the appropriate therapeutic approaches in the past several years. The range of potential treatment regimens extends from judicious observation to early enucleation, with a multitude of variations in between, including teletherapy and brachytherapy.

We chose to utilize iodine 125 in the treatment of selected choroidal melanomas because of its advantageous physical characteristics when compared with other currently popular isotopes, such as cobalt 60. The low-energy roentgen rays from the titanium-encapsulated iodine 125 allow shielding with a thin plate of gold. Thus, the operator and the tissues behind the plaque are protected. The design of the plaque allows adequate irradiation of tumors with thicknesses up to 10 mm. The patients in this report who were followed up for more than two years were treated with gold-backed plaques, without a rim or standardized spacer between the seeds and the sclera. In these plaques, the seeds were placed in a hardened acrylic gel, and the actual distance between the seeds and the sclera was not precisely quantitated, although an attempt was made to keep this distance at approximately 1 mm. The plaques that are in current use have been redesigned utilizing a 1-mm spacer between the seeds and the sclera, which reduces the scleral radiation to doses comparable with those delivered by cobalt 60.[14] Because of its low energy, some investigators have questioned the usefulness of iodine 125 in the treatment of large tumors (tumor thickness of 5 mm or more). The efficacy of iodine 125 has been verified by measurable reduction in tumor mass occurring in 25 of 26 patients in our series, including all nine patients with large tumors. In all nine patients with large tumors, a measurable reduction in tumor height occurred, as measured by A-scan ultrasonography, with an average decrease in tumor thickness of 49% during the study period.

The key to any treatment regimen must be patient survival, as this is one of the few potentially fatal diseases dealt with by the ophthalmologist. Five-year survival for choroidal melanoma treated by enucleation has been reported to be 53% to 96% and is extremely variable depending on tumor size, location, and histologic cell type.[15-17] Similar survival rates have been achieved with local brachytherapy, as reported by other investigators.[3,18-20] To date, two of the 26 patients in our series treated with iodine 125 have died. One patient had biopsy-proved metastatic melanoma to the liver and died of secondary complications. The other patient died of adenocarcinoma of the lung, metastatic to the brain. Although our initial data appear in line with previously reported figures (92% survival to date), further follow-up will be necessary to properly evaluate the eventual mortality.

In addition to survival, retention of useful vision has been the goal of various radiotherapists and has been a motivating force in the development of local brachytherapy for the treatment of malignant choroidal melanomas. In our series, 14 (54%) of 26 patients retained visual acuity within two lines of preoperative vision after a mean follow-up of nearly four years. This is very similar to the series reported by Packer et al[21] in which 16 of 29 patients treated with iodine 125 retained vision within two lines of their preoperative level, at a mean follow-up of 38 months. Of the 12 patients (46%) in our series with greater than two lines of visual acuity loss, significant radiation retinopathy and/or optic neuropathy occurred in eight patients (31%). Radiation retinopathy is characterized by vascular sheathing, telangiectatic vessels, including neovascularization, ischemic infarcts, retinal edema, and exudates. Although it is possible that cystoid macular edema may be caused by inflammation secondary to radiation-induced tumor necrosis, for the purposes of our review, we concluded that all cases of macular edema were radiation induced. In this article, we refer only to clinically visible radiation retinopathy, recognizing that more subtle changes can be demonstrated with fluorescein angiography. Radiation-induced optic neuropathy is characterized by vascular congestion, with or without edema of the optic nerve head and disc pallor. Peripapillary hemorrhages, exudates, and subretinal fluid may be present. Further discussions of radiation retinopathy and optic neuropathy induced by cobalt plaque or external beam irradiation are available in reports published by Brown and coworkers.[22,23]

The risk of visual loss increased with proximity of the tumor to the optic nerve and fovea. Eighteen of our patients had posteriorly located tumors within 3 disc diameters of the fovea and/or optic nerve. Radiation damage to the nerve and/or retina accounted for greater than two lines of visual loss in seven (38%) of these patients. To date, useful visual acuity of 20/200 or greater has been retained in 15 (58%) of 26 of our patients followed up for a mean interval of 45 months. The onset of significant retinopathy and/or optic neuropathy with visual loss was noted to develop at an average of 32 months after treatment. Although the average onset of radiation retinopathy and/or optic neuropathy at 32 months after plaque placement may be delayed when compared with other currently popular radiation modalities,[2,24] the number of patients in our series is small, and it is not possible to precisely compare our data with those published by other investigators, as there is not strict standardization in methods of reporting data.

Most reported series suffer because of relatively small numbers of patients, short follow-up, lack of prospectivity, and absence of randomized, and hence truly comparable, controls. Ophthalmologists, radiation oncologists, epidemiologists, and biostatisticians agree that survival is the paramount issue for patients with eye cancer. Only a prospective randomized clinical trial can definitively answer the question of survival in patients treated with radiation or enucleation. Fortunately, such a trial is now in progress.[25]

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